

Log3: 26th May 2026 (12.30am)

Ada vs Ampere Server Documentation

Scope: This document is based only on the two latest logs:

- `ada_2026-05-25_19-30.log`
- `ampere_2026-05-26_00-42.log`

Earlier checks, earlier conclusions, and previous assumptions are intentionally ignored.

1. Executive Summary

Recommended primary server from these logs: Ada

Ada is the better default server for our Unitree Go2 / Gaussian Splatting / SLAM work when we need GPU compute, large RAM, and high local storage capacity. It has:

- 3 × NVIDIA RTX 6000 Ada Generation GPUs, each with ~48 GiB VRAM.
- 64 logical CPU threads from an Intel Xeon w7-3565X.
- 503 GiB system RAM.
- A large 6.9 TiB NVMe RAID0 root filesystem and a 9.1 TiB `/scratch1`.
- Much lower system load during the test than Ampere.

Ampere is still useful, but the current log shows it was heavily loaded:

- 2 × RTX A6000 GPUs, each ~48 GiB VRAM.
- GPUs were already under high memory and compute load.
- CPU run queue/load was very high.
- Only 125 GiB RAM.
- Native OS is Ubuntu 22.04, which is convenient for ROS 2 Humble-style workflows, but ROS 2/Gazebo were not installed or not visible in the user PATH at the time of testing.

Practical decision

Ada should still be treated as the main compute server for the Unitree Go2 project, especially for GPU-heavy Gaussian Splatting, CUDA/PyTorch experiments, dataset processing, and large benchmarking work.

However, the latest checks show that the intended Ada scratch workspace is not currently writable by user e21283. Creating `/scratch1/e21283` failed with "Permission denied". Therefore, Ada is the best primary server only after the scratch workspace permission issue is fixed.

Ampere remains useful as a secondary/fallback server, especially because it runs Ubuntu 22.04, which is more convenient for ROS 2 Humble-style workflows. But the latest Ampere check showed both RTX A6000 GPUs were heavily occupied in memory, and `/storage/scratch2` was also not writable by user e21283. So Ampere should only be used after checking GPU availability and confirming a writable scratch workspace.

Main updated decision:

Use Ada as the primary compute target.

Use Ampere as the secondary/fallback target.

Do not begin heavy project work on either server until a writable scratch directory is confirmed.

2. Host Identity and Network

Item	Ada	Ampere
Hostname	ada	ampere
Main IP observed	10.40.18.12	10.40.18.10
Extra IPs	172.17.0.1 Docker bridge also present	none shown
Log start time	2026-05-25 19:30:45 +00:00	2026-05-26 00:42:52 +05:30
Tested user	e21283	e21283

Network checks from Ampere

Ampere could reach the internet:

```
ping google.com
9 packets transmitted, 9 received, 0% packet loss
avg RTT ≈ 44.006 ms
```

Ampere could reach Ada very quickly on the internal network:

```
ping ada.ce.pdn.ac.lk
8 packets transmitted, 8 received, 0% packet loss
avg RTT ≈ 0.355 ms
```

Ampere loopback/self ping to 10.40.18.10 was also successful:

```
14 packets transmitted, 14 received, 0% packet loss
avg RTT ≈ 0.036 ms
```

DNS note

The latest access attempt from Tesla showed that:

```
ssh -X e21283@ada.ce.pdn.ac.lk
```

failed with:

```
ssh: Could not resolve hostname ada.ce.pdn.ac.lk: Name or service not known
```

However, IP-based access worked. The observed working access path was:

Tesla → Ampere using 10.40.18.10

Ampere → Ada using 10.40.18.12

So for now, IP-based SSH should be preferred when DNS names fail:

Ampere: `ssh -X e21283@10.40.18.10`

Ada: ssh -X e21283@10.40.18.12

The latest checks also confirmed the user and group on both servers as:

user: e21283

group: e21

3. Operating System and Kernel

Item	Ada	Ampere
OS	Ubuntu 24.04.3 LTS	Ubuntu 22.04.5 LTS
Codename	noble	jammy
Kernel	6.8.0-117-generic	6.8.0-111-generic
Architecture	x86_64	x86_64

Interpretation

Ada has a newer OS. That is good for newer compilers and system packages, but can be inconvenient if the project specifically targets ROS 2 Humble, which is normally easiest on Ubuntu 22.04.

Ampere has Ubuntu 22.04, which is more aligned with a ROS 2 Humble/Gazebo Classic-style setup. However, in the current log, `ros2`, `gazebo`, and `gz` were not found in the visible PATH.

4. Current Server Load at Test Time

Item	Ada	Ampere
Uptime	~5 days, 7 hours	~5 days, 6 hours
Logged-in users	18	20
Load average	~6.4, 6.5, 7.0	~43.6, 49.6, 49.0
CPU idle from <code>vmstat</code> samples	mostly ~89% after first sample	often only ~11-27% idle
Overall state	Moderate load	Heavy load

Interpretation

Ada was much less loaded at the time of testing. Ampere was under strong active CPU/GPU load, so performance tests on Ampere during this window would not represent its best capability.

5. CPU Hardware

Item	Ada	Ampere
CPU model	Intel Xeon w7-3565X	Intel Xeon Silver 4215R
Logical CPUs	64	32
Sockets	1	2
Cores/socket	32	8
Threads/core	2	2
Max frequency	4.8 GHz	4.0 GHz

Item	Ada	Ampere
NUMA nodes	1	2
L3 cache	82.5 MiB	22 MiB total across 2 instances

Interpretation

Ada has a stronger and simpler CPU platform:

- More logical threads.
- More cores.
- Higher boost clock.
- Larger L3 cache.
- Single NUMA node, which simplifies CPU/GPU memory locality.

Ampere has a dual-socket NUMA design. This is not bad, but for GPU-heavy workloads it can add locality issues, especially if processes are scheduled on the CPU socket farther from the GPUs.

6. System Memory

Item	Ada	Ampere
RAM total	503 GiB	125 GiB
RAM used at check	188 GiB	36 GiB
RAM available	314 GiB	85 GiB
Swap total	8 GiB	127 GiB
Swap used	6.7 GiB	293 MiB

Interpretation

Ada has a major advantage for:

- Large datasets.
- Large point clouds.
- Gaussian scene preprocessing.
- Multiple parallel experiments.
- CPU-side training/preprocessing steps.

Ada had high swap usage, but still had 314 GiB RAM available. This suggests old or long-running jobs may have paged memory out, but the machine still had a lot of memory headroom.

Ampere has much larger swap but much less RAM. Heavy 3DGS/SLAM preprocessing jobs are safer on Ada.

7. GPU Hardware and Driver State

Latest Ada GPU state

Driver Version: 580.82.07

CUDA Version shown by nvidia-smi: 13.0

GPUs: 3 × NVIDIA RTX 6000 Ada Generation

VRAM per GPU: 49140 MiB

Latest observed Ada GPU usage:

GPU 0: RTX 6000 Ada Generation, 13017 MiB / 49140 MiB used, 5% utilization, 37 C

GPU 1: RTX 6000 Ada Generation, 13226 MiB / 49140 MiB used, 0% utilization, 45 C

GPU 2: RTX 6000 Ada Generation, 12603 MiB / 49140 MiB used, 0% utilization, 38 C

Interpretation:

Ada GPUs are not empty, but they are much more available than Ampere in the latest checks. Each GPU already had around 12.6–13.2 GiB VRAM occupied, leaving roughly 35–36 GiB free per GPU at that moment.

Latest Ampere GPU state

Driver Version: 550.144.03

CUDA Version shown by nvidia-smi: 12.4

GPUs: 2 × NVIDIA RTX A6000

VRAM per GPU: 49140 MiB

Latest observed Ampere GPU usage:

GPU 0: RTX A6000, 47166 MiB / 49140 MiB used, 100% utilization, 87 C

GPU 1: RTX A6000, 36723 MiB / 49140 MiB used, 0% utilization, 81 C

Interpretation:

Ampere GPU 0 was effectively unavailable because it was nearly full and running at 100% utilization.

Ampere GPU 1 had 0% compute utilization at the instant of checking, but it still had 36723 MiB of VRAM already occupied. That means it was also not actually free for large 3DGS or SLAM experiments.

Updated project interpretation:

Ada remains the better GPU target because it has three newer RTX 6000 Ada Generation GPUs and significantly more available GPU memory during the latest check.

Ampere can still run GPU work, but it must be checked immediately before use with nvidia-smi. Based on the latest output, Ampere was not a good immediate choice for heavy training.

GPU topology

Item	Ada	Ampere
GPU interconnect shown	NODE between GPUs	NODE between GPUs
NVLink shown	No	No
GPU CPU affinity	0–63	0–7,16–23
NUMA affinity	0	0

8. Graphics, X11, OpenGL, RViz, and GUI State

Ada

```
OpenGL vendor string: Mesa
OpenGL renderer string: llvmpipe (LLVM 20.1.2, 256 bits)
glxgears: 16 frames in 5.1 seconds = 3.121 FPS
rviz2: command not found
xclock: launches, but with a charset warning
```

Ampere

```
OpenGL vendor string: Mesa
OpenGL renderer string: llvmpipe (LLVM 15.0.7, 256 bits)
glxgears: 3.5–8.4 FPS
rviz2: command not found
xclock: launches, but with a charset warning
```

Interpretation

Both servers are using software OpenGL rendering through llvmpipe for the tested X-forwarded GUI session. That means RViz/Gazebo GUI rendering over X11 will be slow and not hardware-accelerated in the tested configuration.

This does not prevent headless ROS/Gazebo/SLAM work, but it does mean:

- Do not judge GPU rendering capability from `glxgears`.
- Use headless Gazebo where possible.
- Use local visualization on your laptop if needed.
- For RViz, consider running RViz locally and streaming ROS topics, or use X forwarding only for light debugging.

9. Storage Layout

9.1 Ada Storage

`df -h` showed:

Mount	Size	Used	Available	Use%	Type / Meaning
<code>/</code>	6.9T	4.7T	2.0T	71%	local NVMe RAID0 root
<code>/scratch1</code>	9.1T	2.2T	6.4T	26%	local scratch disk
<code>/new-home</code>	28T	296G	28T	2%	NFS student-home
<code>/home</code>	11T	11T	50G	100%	NFS home, full

`lsblk` showed:

- Two 3.6T NVMe disks combined into `/dev/md126`, mounted as `/`.
- One 9.1T disk mounted as `/scratch1`.
- Another 9.1T disk exists but was not shown mounted in the log.

Ada storage recommendation

The intended heavy-work location on Ada is still:

`/scratch1/e21283/`

This should be used for datasets, ROS bags, build directories, Gaussian Splatting outputs, and temporary experiments once permission is available.

However, the latest check showed that user e21283 currently cannot create this directory:

```
mkdir -p /scratch1/e21283/unitree_go2/test_io
```

failed with:

```
mkdir: cannot create directory '/scratch1/e21283': Permission denied
```

Because that mkdir failed, the following cd also failed:

```
cd /scratch1/e21283/unitree_go2/test_io
```

The dd test that followed did not run inside /scratch1. It ran from the current directory, which was still /home/e21283.

The observed dd result was:

3.0 GiB copied in 316.124 s, 10.2 MB/s

This is not a valid scratch disk benchmark. It is only a measurement of the current fallback write path, most likely the NFS home directory.

Updated conclusion:

/scratch1 exists on Ada, but /scratch1/e21283 is not currently writable or creatable by user e21283. Admin help or the correct assigned scratch path is needed before heavy work should be started on Ada.

9.2 Ampere Storage

`df -h` showed:

Mount	Size	Used	Available	Use%	Type / Meaning
<code>/</code>	468G	205G	240G	47%	local NVMe root
<code>/storage/datasets</code>	2.0T	1.1T	795G	59%	local dataset partition
<code>/storage/scratch1</code>	1.7T	1.1T	521G	67%	local scratch partition
<code>/storage/scratch2</code>	2.0T	417G	1.5T	22%	local scratch partition
<code>/storage/scratch3</code>	1.7T	1.2T	368G	77%	local scratch partition
<code>/new-home</code>	28T	296G	28T	2%	NFS student-home
<code>/home</code>	11T	11T	50G	100%	NFS home, full
<code>/home2</code>	11T	11T	50G	100%	NFS home2, full

Correct scratch paths on Ampere are:

```
/storage/scratch1  
/storage/scratch2  
/storage/scratch3
```

Incorrect paths attempted in the log:

```
/storage/scratch  
/scratch1
```

These do not exist on Ampere.

Ampere storage recommendation

The best intended scratch location on Ampere is still:

```
/storage/scratch2/e21283/
```

because /storage/scratch2 had the most free space in the earlier storage check.

However, the latest check showed that user e21283 currently cannot create this directory:

```
cd /storage/scratch2
```

```
mkdir -p e21283/unitree_go2/test_io
```

failed with:

```
mkdir: cannot create directory 'e21283': Permission denied
```

A direct write test inside /storage/scratch2 also failed:

```
dd: failed to open 'test_4G.img': Permission denied
```

Updated conclusion:

/storage/scratch2 exists on Ampere, but user e21283 cannot currently create /storage/scratch2/e21283 or write files directly inside /storage/scratch2. Admin help or the correct assigned scratch path is needed before heavy work should be started on Ampere.

10. NFS Home and Project Data Policy

Both machines mount the same or related NFS home infrastructure. `/home` was effectively full on both:

```
/home: 11T size, 11T used, 50G available, 100%
```

This matters because many commands default to writing in `~/`, including:

- source builds,
- conda environments,
- pip caches,
- ROS workspaces,
- colcon build logs,
- datasets,

- model checkpoints,
- Gaussian `.ply` / `.splat` files,
- ROS bag recordings.

Rule for this project

Do not use `/home/e21283` for heavy work.

Recommended pattern:

The intended project workspaces are:

Ada:

`/scratch1/e21283/unitree_go2/`

Ampere:

`/storage/scratch2/e21283/unitree_go2/`

But the latest checks show that both intended locations currently fail with permission errors.

Therefore, these commands should not be treated as working yet:

Ada:

`mkdir -p /scratch1/e21283/unitree_go2/{src,data,build,logs,models,rosbags}`

Ampere:

`mkdir -p /storage/scratch2/e21283/unitree_go2/{src,data,build,logs,models,rosbags}`

The correct policy is:

Do not use `/home/e21283` for heavy work.

Do not start large builds, datasets, ROS bags, or model outputs until a writable scratch directory is confirmed.

Ask admin to create the per-user scratch directory, or ask for the correct assigned scratch path for e21283.

Use home only for:

SSH keys

shell config

small notes

small scripts

symlinks pointing into scratch

11. Installed / Missing Tools

Latest confirmed tool state:

nvidia-smi:

Ada: available

Ampere: available

nvcc:

Ada: missing / not visible from earlier checks

Ampere: missing / not visible from earlier checks

ROS 2:

Ada: not found

Ampere: not found / not visible

Gazebo:

Ada: not found

Ampere: not found / not visible

gz:

Ada: not found / not visible

Ampere: not found / not visible

rviz2:

Ada: missing from earlier checks

Ampere: missing from earlier checks

Singularity:

Ada: /usr/bin/singularity exists, but it reports Apptainer version 1.4.5

Ampere: singularity-ce version 3.9.7-bionic

Apptainer:

Ada: /usr/bin/apptainer, version 1.4.5

Ampere: missing from earlier checks

Docker:

Ada: not confirmed usable

Ampere: missing from earlier checks

Python:

Ada: Python 3.12.3

Ampere: Conda Python 3.13.11 from earlier checks

pip:

Ada: pip 24.0

Ampere: pip 25.3 from earlier checks

conda:

Ada: command not found

Ampere: conda 25.11.1 from earlier checks

PyTorch:

Ada: not checked in latest Ada log

Ampere: not installed in base conda environment from earlier checks

Interpretation

Ada now has the better container situation because Apptainer 1.4.5 is available. The singularity command also exists on Ada, but it reports Apptainer, so it should be treated as a Singularity-compatible Apptainer setup.

Ampere has Singularity CE, which is still useful for containers, but Apptainer was not visible in earlier checks.

Neither server currently exposes ROS 2, Gazebo, gz, or RViz2 in the tested user environment.

Ada does not currently have conda visible. Ampere does have conda, but PyTorch was not installed in the base environment.

Neither server is immediately ready for the full Unitree Go2 project stack without setting up a controlled environment.

Best approach:

Use Apptainer/Singularity for reproducible ROS/Gazebo environments.

Use conda/mamba only where available, or install a user-local Python environment.

Avoid relying on the system base environment.

Record every environment setup command carefully.

12. Containers and Admin Restrictions

Containers and Admin Restrictions

Ada has Apptainer installed:

which apptainer

/usr/bin/apptainer

apptainer --version

apptainer version 1.4.5

Ada also has a singularity command:

which singularity

/usr/bin/singularity

singularity --version

apptainer version 1.4.5

This means Ada can likely run Singularity/Apptainer-style containers, with singularity behaving as a compatibility command for Apptainer.

Ampere has Singularity CE:

singularity-ce version 3.9.7-bionic

From earlier checks, Ampere did not have Docker or Apptainer visible.

The latest group check on both servers showed only:

e21

No docker/admin-style group was shown in the provided logs.

Practical meaning:

Use Apptainer on Ada.

Use Singularity CE on Ampere.

Do not assume Docker.

Do not assume sudo/admin privileges.

For ROS 2 Humble and Gazebo, a container-based environment is likely the cleanest route unless the admin installs the native stack.

13. ROS 2 / Gazebo Readiness

ROS 2 / Gazebo Readiness

From the latest checks:

Ada:

ros2 was not found.

ros2 --version returned command not found.

gazebo --version returned command not found.

which gazebo returned nothing.

which gz returned nothing.

Ampere:

which ros returned nothing.

which ros2 returned nothing.

which gazebo returned nothing.

which hz returned nothing.

which gz returned nothing.

rviz2 was also missing in the earlier checks.

Interpretation:

Neither Ada nor Ampere is currently ready as a native ROS 2 / Gazebo workstation for user e21283.

Ampere is still more naturally aligned with ROS 2 Humble because it runs Ubuntu 22.04, but ROS 2 and Gazebo are not currently visible in the user PATH.

Ada has stronger hardware and Apptainer available, so it may be a better target for a containerized ROS/Gazebo or 3DGS workflow.

Recommended options:

1. Use an Apptainer/Singularity ROS 2 Humble container.
This is especially useful on Ada because Ada runs Ubuntu 24.04 while many ROS 2 Humble workflows expect Ubuntu 22.04.
2. Ask admin to install ROS 2 Humble and Gazebo.
This is more natural on Ampere because it is already Ubuntu 22.04.
3. Split the workflow.
Use Ada for GPU-heavy 3DGS and large-data processing.
Use a local laptop or containerized Ubuntu 22.04 environment for ROS/Gazebo simulation.
Move ROS bags, point clouds, and outputs between environments.

14. GUI and Visualization Strategy

The logs show X forwarding works enough to launch simple X apps like `xclock`, but OpenGL uses CPU software rendering.

Recommended visualization practice

For project work:

- Use servers for compute, training, bag processing, and headless simulation.
- Avoid heavy remote GUI rendering.
- Use local visualization when possible:
 - RViz locally.

- Foxglove Studio locally.
- browser-based tools locally.
- copy `.ply`, `.splat`, `.bag`, `.db3`, and logs from server to laptop for inspection.

For 3D Gaussian Splatting outputs, use browser viewers locally unless server-side remote visualization is specifically required.

15. Benchmarking Quality of Current Tests

Good checks already completed

The logs successfully established:

- hostname and IPs,
- OS/kernel versions,
- CPU model and topology,
- RAM/swap state,
- GPU model/driver/usage,
- basic GPU topology,
- OpenGL/X-forwarding state,
- storage mounts,
- NFS status,
- some missing tools,
- Ampere internet/internal connectivity,
- Ampere Python/conda/PyTorch state.

Incomplete or unreliable checks

Real scratch speed:

Not measured. Both Ada and Ampere scratch write attempts failed due to permission errors.

Ada dd speed:

Not a valid scratch benchmark. The `/scratch1` cd failed, so dd ran from `/home/e21283` and showed about 10.2 MB/s.

Ampere dd speed:

Not measured. Writing to `/storage/scratch2/test_4G.img` failed with permission denied.

Scratch permissions:

Now confirmed as a problem. `/scratch1/e21283` on Ada and `/storage/scratch2/e21283` on Ampere could not be created by user `e21283`.

Python GPU framework:

Ampere base has no torch from earlier checks. Ada PyTorch state has not been checked.

CUDA toolkit:

nvcc is missing or not visible from earlier checks. Full CUDA development library state is still not confirmed.

ROS 2:

Not visible on either server.

Gazebo:

Not visible on either server.

Apptainer/Singularity:

Ada has Apptainer 1.4.5. Ampere has Singularity CE 3.9.7-bionic.

Correct assigned scratch path:

Still unknown. Need admin confirmation.

Long-term stability:

Not tested.

Real compile performance:

Not tested.

Multi-GPU training behavior:

Not tested.

GUI RViz performance:

Not tested because RViz is not installed and OpenGL forwarding uses llvmpipe.

16. Server-by-Server Assessment

16.1 Ada Assessment

Strengths

- Best GPU hardware: 3 × RTX 6000 Ada Generation.
- Best CPU: Xeon w7-3565X, 64 logical CPUs.
- Best RAM: 503 GiB.
- Best local working space: 6.9T root plus 9.1T `/scratch1`.
- Lower load than Ampere during the test.
- Newer NVIDIA driver: 580.82.07.
- Good for large-scale 3DGS and GPU-heavy benchmarking.

Weaknesses / risks

- Ubuntu 24.04 may complicate native ROS 2 Humble workflows.
- nvcc is not visible from earlier checks.
- ROS 2 is not installed or not visible.
- Gazebo is not installed or not visible.
- RViz2 is not installed or not visible.
- conda is not installed or not visible.
- OpenGL over X forwarding is software-rendered.
- Creating `/scratch1/e21283` failed with permission denied.

- The latest dd write ran from /home/e21283, not scratch, so it is not a valid scratch benchmark.
- Home storage is full/slow and should not be used for heavy work.
- GPUs are not completely empty; around 12.6–13.2 GiB VRAM was already used per GPU in the latest check.

Best use

Use Ada for:

- Gaussian Splatting training.
 - CUDA/PyTorch experiments.
 - map processing,
 - large ROS bag post-processing,
 - evaluation metrics,
 - paper-quality benchmarking,
 - storage-heavy experiments, but only after /scratch1/e21283 or another writable scratch path is confirmed.
-

16.2 Ampere Assessment

Strengths

- Ubuntu 22.04 matches many ROS 2 Humble workflows better than Ada.
- 2 × RTX A6000 GPUs with 48 GiB VRAM each.
- Singularity CE installed.
- Multiple local storage partitions under `/storage`.
- Internet and internal network connectivity verified.
- Conda installed for user `e21283`.

Weaknesses / risks

- Heavy load was observed in earlier checks.
- Latest GPU 0 was almost full and running at 100% utilization.
- Latest GPU 1 had 36723 MiB VRAM already occupied even though utilization was 0% at that instant.
- Only 125 GiB RAM from earlier detailed checks.
- Dual-socket NUMA system can be more complicated.
- nvcc is not visible.
- PyTorch is not installed in the base conda environment.
- ROS 2, Gazebo, gz, and RViz2 are not installed or not visible.
- Docker and Apptainer are missing from earlier checks.

- Home and home2 are full.
- Creating /storage/scratch2/e21283 failed with permission denied.
- Writing directly to /storage/scratch2/test_4G.img failed with permission denied.
- Scratch path differs from Ada; /scratch1 does not exist here.

Best use

Use Ampere for:

- fallback GPU jobs when Ada is unavailable,
- containerized ROS 2 work if Singularity images are prepared,
- native Ubuntu 22.04 experiments,
- smaller GPU experiments,
- jobs that do not require the full 503 GiB RAM available on Ada.

17. Recommended Project Policy From Now On

Primary compute server

Use:

```
ada
```

as the default server for this project.

- Primary Ada workspace — intended, but not currently writable
- The intended Ada workspace is:
 - /scratch1/e21283/unitree_go2/
- But the latest check showed that /scratch1/e21283 cannot currently be created by user e21283.
- Do not assume this path is usable until permission is fixed.
- Secondary Ampere workspace — intended, but not currently writable
- The intended Ampere workspace is:
 - /storage/scratch2/e21283/unitree_go2/
- But the latest check showed that /storage/scratch2/e21283 cannot currently be created by user e21283.
- Do not assume this path is usable until permission is fixed.
- Current workspace rule
- Before running builds, datasets, ROS bags, or model outputs, first confirm a writable scratch path.
- The next practical task is not benchmarking. The next practical task is to get the correct scratch directory created or confirmed by admin.

Do not run heavy jobs in home

Avoid:

```
/home/e21283
```

for heavy work.

Before every GPU run

Always check:

```
hostname  
nvidia-smi  
free -h  
uptime  
df -h
```

Then choose a GPU explicitly:

```
export CUDA_VISIBLE_DEVICES=0
```

or whichever GPU is free.

For long jobs

Use `tmux`, but avoid nested tmux mistakes.

Correct command to escape nested tmux state:

```
unset TMUX  
tmux new -s unitree
```

The log used `unset $TMUX`, which is not the correct form. The correct variable name is `TMUX`, without `$`.

18. Immediate Next Commands to Run

Immediate Next Commands to Run

The previous scratch benchmark commands failed because the directories were not writable. The next checks should focus on confirming permissions and finding the correct writable workspace.

On Ada:

```
hostname  
whoami  
groups  
pwd
```

```
ls -ld /scratch1
```

```
ls -ld /scratch1/e21283
```

```
touch /scratch1/test_write_e21283
touch /scratch1/e21283/test_write_e21283

df -h /scratch1
nvidia-smi
```

Expected based on current logs:

/scratch1 exists.
/scratch1/e21283 is not currently creatable or writable by user e21283.
The touch commands will likely fail unless permissions have been changed.

On Ampere:

```
hostname
whoami
groups
pwd

ls -ld /storage/scratch1
ls -ld /storage/scratch2
ls -ld /storage/scratch3
ls -ld /storage/scratch2/e21283
touch /storage/scratch2/test_write_e21283
touch /storage/scratch2/e21283/test_write_e21283

df -h /storage/scratch2
nvidia-smi
```

Expected based on current logs:

/storage/scratch2 exists.
/storage/scratch2/e21283 is not currently creatable or writable by user e21283.
Writing directly inside /storage/scratch2 will likely fail unless permissions have changed.

After scratch access is fixed, rerun the actual disk test from the confirmed writable scratch directory.

19. Final Decision

Based only on the latest provided checks:

Ada should still be selected as the main server for the Unitree Go2 project, especially for GPU-heavy Gaussian Splatting, CUDA/PyTorch experiments, large-data processing, and benchmarking.

Ampere should remain a secondary option, mainly for Ubuntu 22.04-compatible workflows, Singularity-based ROS 2 Humble environments, or overflow GPU jobs.

However, the immediate blocker is not server choice anymore. The immediate blocker is scratch workspace access.

Current blocker:

Ada: /scratch1/e21283 could not be created due to permission denied.

Ampere: /storage/scratch2/e21283 could not be created due to permission denied.

Therefore, before starting real project work:

Get a writable scratch directory confirmed on Ada.

Get a writable scratch directory confirmed on Ampere if Ampere will be used.

Do not run heavy jobs from /home/e21283.

After scratch access is fixed, rerun the disk write benchmark from the correct scratch location.

Then choose the active server based on nvidia-smi, free -h, uptime, and df -h at the time of use.

Final operational decision:

Primary server: Ada, after scratch access is fixed.

Secondary server: Ampere, only when GPU memory is available and scratch access is fixed.

Most urgent next task: resolve scratch directory permissions.